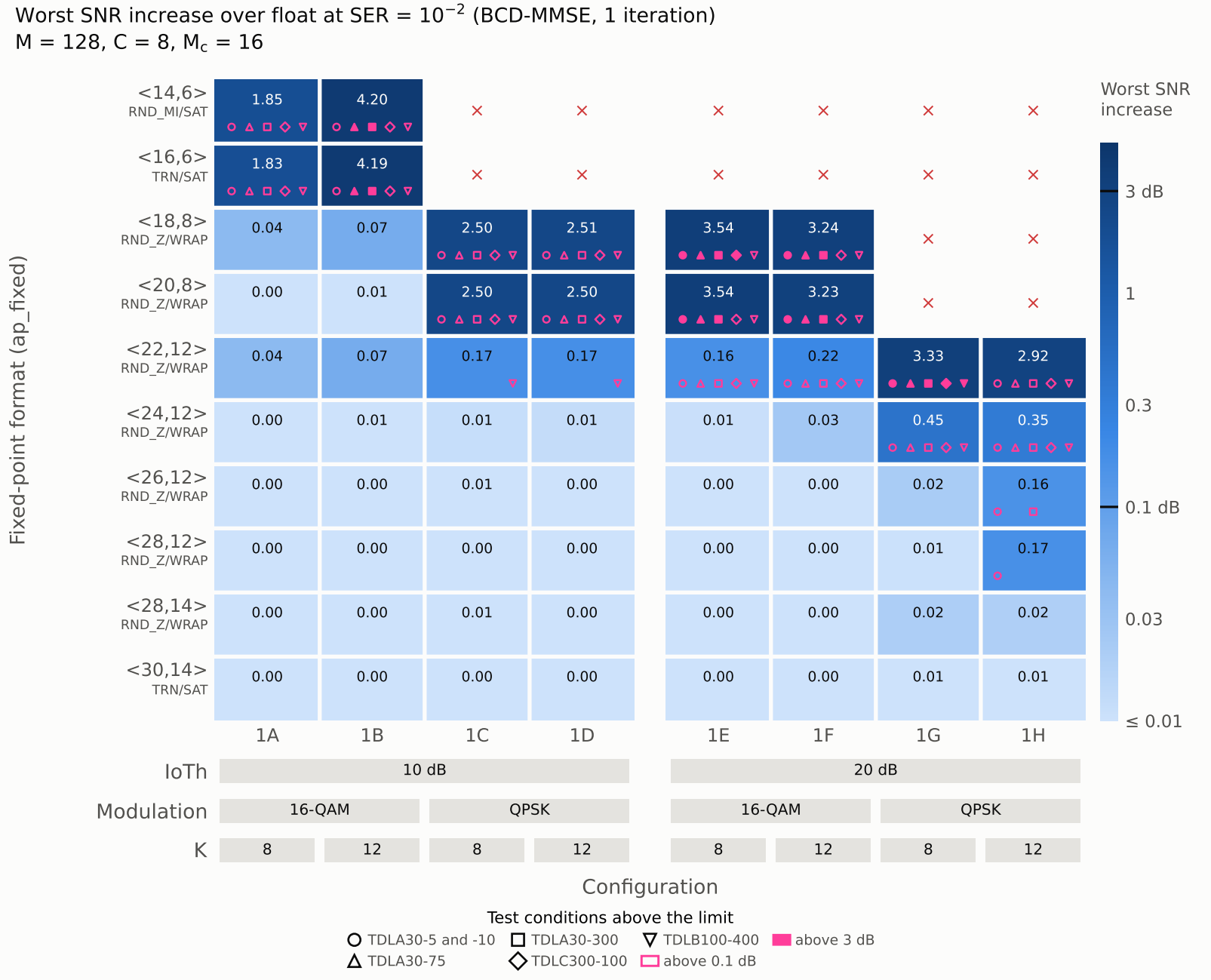
Worst SNR increase over float at SER = 10^{-2} (BCD-MMSE, 1 iteration)
M = 128, C = 8, M_c = 16
Fixed-point format (ap_fixed)
<14,6> RND_MI/SAT
<16,6> TRN/SAT
<18,8> RND_Z/WRAP
<20,8> RND_Z/WRAP
<22,12> RND_Z/WRAP
<24,12> RND_Z/WRAP
<26,12> RND_Z/WRAP
<28,12> RND_Z/WRAP
<28,14> RND_Z/WRAP
<30,14> TRN/SAT
1.85
4.20
1.83
4.19
0.04 0.07 2.50 2.51 3.54 3.24
0.00 0.01 2.50 2.50 3.54 3.23
0.04 0.07 0.17 0.17 0.16 0.22 3.33 2.92
0.00 0.01 0.01 0.01 0.01 0.03 0.45 0.35
0.00 0.00 0.01 0.00 0.00 0.00 0.02 0.16
0.00 0.00 0.00 0.00 0.00 0.00 0.01 0.17
0.00 0.00 0.01 0.00 0.00 0.00 0.02 0.02
0.00 0.00 0.00 0.00 0.00 0.00 0.01 0.01
1A 1B 1C 1D 1E 1F 1G 1H
IoTh
10 dB
20 dB
Modulation
16-QAM
QPSK
16-QAM
QPSK
K
8 12 8 12 8 12 8 12
Configuration
Worst SNR increase
3 dB
1
0.3
0.1 dB
0.03
≤ 0.01
Test conditions above the limit
TDLA30-5 and -10
TDLA30-75
TDLA30-300
TDLC300-100
TDLB100-400
above 0.1 dB
above 3 dB